Linear momentum

$\vec{S} = cW\hat{n}$ but $p = \frac{\mathcal{E}}{c} = \frac{S}{c^2} \Rightarrow \vec{P} = \frac{1}{c^2} \frac{\vec{E}\times\vec{B}}{\mu_0} = \varepsilon_0 \vec{E}\times\vec{B}$

$$\Rightarrow \vec{P} = \frac{\varepsilon_0}{2}\int d\vec{r}\left(\vec{E}\times\vec{B} - \vec{B}\times\vec{E}\right) \rightarrow \sum_{\vec{k},\sigma} \hbar\vec{k}\left(\hat{a}^{+}_{k\sigma}\hat{a}_{k\sigma} + \frac{1}{2}\right)$$

$$= \sum_{\vec{k},\sigma} \hbar\vec{k}\,\hat{a}^{+}_{k\sigma}\hat{a}_{k\sigma} \; !$$

(calculational steps are similar to angular momentum which is considered next.)

Angular momentum of EM field

$$\vec{J}(\vec{r}_0) = \varepsilon_0 \int (\vec{r}-\vec{r}_0) \times \left(\vec{E}(\vec{r},t) \times \vec{B}(\vec{r},t)\right) d^3\vec{r}$$

with respect to the point $\vec{r}_0$.

$$\vec{J}(\vec{r}_0) = \int (\vec{r}-\vec{r}_0) \times \vec{P}\, d^3\vec{r} = \vec{J}(0) - \vec{r}_0 \times \vec{P}$$

Rewrite using $\vec{B} = \vec{\nabla} \times \vec{A}$, $B_\ell = \varepsilon_{\ell k p} \nabla_k A_p$

$$\left(\vec{E} \times \vec{B}\right)_n = \varepsilon_{nm\ell} E_m B_\ell = \underbrace{\varepsilon_{nm\ell}\, \varepsilon_{\ell k p}}_{\delta_{nk}\delta_{mp} - \delta_{np}\delta_{mk}} E_m \nabla_k A_p =$$

$$= E_m \nabla_n A_m - E_m \nabla_m A_n \quad , \nabla_n \equiv \frac{\partial}{\partial x_n}$$

Then ℓ-th component of $\vec{J}$, J_ℓ is proportional to

$$\left\{(\vec{r}-\vec{r}_0) \times (\vec{E} \times \vec{B})\right\}_\ell = \varepsilon_{\ell j n} (r_j - r_{0j}) \left(E_m \frac{\partial}{\partial x_n} A_m - E_m \frac{\partial}{\partial x_m} A_n\right)$$

$$= \varepsilon_{\ell j n} (r_j - r_{0j}) E_m \frac{\partial}{\partial x_n} A_m - \varepsilon_{\ell j n} (r_j - r_{0j}) \underbrace{\frac{\partial}{\partial x_m}(E_m A_n)}_{E_m \frac{\partial}{\partial x_m} A_n + \underbrace{\left(\frac{\partial}{\partial x_m} E_m\right)}_{\text{but } \vec{\nabla}\cdot\vec{E} = 0} A_n}$$

(here r_j is the same as x_j, of course)

$$= \varepsilon_{\ell j n} (r_j - r_{0j}) E_m \frac{\partial}{\partial x_n} A_m - \varepsilon_{\ell j n} \frac{\partial}{\partial x_m}\left((r_j - r_{0j}) E_m A_n\right)$$

$$+ \varepsilon_{\ell j n} \underbrace{\left(\frac{\partial}{\partial x_m}(r_j - r_{0j})\right)}_{\delta_{mj}} E_m A_n = E_m \left(\varepsilon_{\ell j n} (r_j - r_{0j}) \frac{\partial}{\partial x_n}\right) A_m$$

$$- \frac{\partial}{\partial x_m}\left(E_m\, \varepsilon_{\ell j n} (r_j - r_{0j}) A_n\right)$$

$$+ \varepsilon_{\ell m n} E_m A_n .$$

$$J_\ell(\vec{r}_0) = \varepsilon_0 \int d^3\vec{r} \Big\{ E_m \left[(\vec{r}-\vec{r}_0) \times \vec{\nabla} \right] A_m \underbrace{- \vec{\nabla} \cdot \vec{E} \left[(\vec{r}-\vec{r}_0) \times \vec{A} \right]_\ell}_{} + (\vec{E} \times \vec{A})_\ell \Big\}$$

this reduces to the surface contribution which vanishes at spatial infinity.

$$\Rightarrow \vec{J}(\vec{r}_0) = \vec{J}_{orb}(\vec{r}_0) + \vec{J}_{spin}$$

orbital angular momentum $\vec{J}_{orb}(\vec{r}_0) = \varepsilon_0 \int dV\, E_m \left[(\vec{r}-\vec{r}_0) \times \vec{\nabla} \right] A_m$

spin (intrinsic) angular momentum $= \vec{J}_{orb}(0) - \vec{r}_0 \times \vec{P}$.

$$\vec{J}_{spin} = \varepsilon_0 \int d^3\vec{r} \left(\vec{E}(\vec{r},t) \times \vec{A}(\vec{r},t) \right)$$

(independent of $\vec{r}_0$!)

$$\vec{A}(\vec{r},t) = \sum_{k\sigma} \sqrt{\frac{\hbar}{2\varepsilon_0 V \omega_k}} \left\{ \hat{a}_{k\sigma} e^{-i(\omega_k t - \vec{k}\cdot\vec{r})} \hat{e}_{k\sigma} + \hat{a}^+_{k\sigma} e^{i(\omega_k t - \vec{k}\cdot\vec{r})} \hat{e}^*_{k\sigma} \right\}$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = \sum_{k\sigma} i \sqrt{\frac{\hbar \omega_k}{2\varepsilon_0 V}} \left\{ \hat{a}_{k\sigma} e^{-i\omega_k t + i\vec{k}\cdot\vec{r}} \hat{e}_{k\sigma} - \hat{a}^+_{k\sigma} e^{i\omega t - i\vec{k}\cdot\vec{r}} \hat{e}^*_{k\sigma} \right\}$$

$$\cancel{\sum_{k\sigma} \sum_{k'\sigma'} \int d\vec{r} \frac{\hbar}{2\varepsilon_0} \sqrt{\frac{\omega_k}{\omega_{k'}}} \Big\{ \hat{a}_{k\sigma} e^{-i\omega_k t + i\vec{k}\cdot\vec{r}} \hat{e}_{k\sigma} - \hat{a}^+_{k\sigma} e^{i\omega_k t - i\vec{k}\cdot\vec{r}} \hat{e}^*_{k\sigma} \Big\} \times \Big\{ \hat{a}_{k'\sigma'} e^{-i\omega'_k t + i\vec{k}'\cdot\vec{r}} \hat{e}_{k'\sigma'} - \hat{a}^+_{k'\sigma'} e^{i\omega'_k t - i\vec{k}'\cdot\vec{r}} \hat{e}^*_{k'\sigma'} \Big\}}$$

Importantly, operators $\vec{E}$ and $\vec{A}$ do not commute!

$\Rightarrow$ quantum expression for $\vec{J}_{spin}$ is obtained by "symmetrization"

$$\hat{\vec{J}}_{spin} = \frac{\varepsilon_0}{2} \int d^3\vec{r} \left(\hat{\vec{E}}(r,t) \times \hat{\vec{A}}(r,t) - \hat{\vec{A}}(r,t) \times \hat{\vec{E}}(r,t) \right)$$

$$\vec{J}_{spin} = \frac{\varepsilon_0}{2} \sum_{k\sigma} \sum_{k'\sigma'} \int d^3\vec{r}\, \frac{\hbar}{2\varepsilon_0 V} \sqrt{\frac{\omega_k}{\omega_{k'}}} \left(i\hat{a}_{k\sigma}\, e^{-i\omega_k t + i\vec{k}\cdot\vec{r}}\, \hat{e}_{k\sigma} - i\hat{a}^{+}_{k\sigma}\, e^{i\omega_k t - i\vec{k}\vec{r}}\, \hat{e}^{*}_{k\sigma} \right)$$

$$\times \left(\hat{a}_{k'\sigma'}\, e^{-i\omega'_k t + ik'r}\, \hat{e}_{k'\sigma'} + \hat{a}^{+}_{k'\sigma'}\, e^{i\omega'_k t - ik'r}\, \hat{e}^{*}_{k'\sigma'} \right)$$

$$- \sqrt{\frac{\omega_{k'}}{\omega_k}} \left(\hat{a}_{k\sigma}\, e^{-i\omega_k t}\, \hat{e}_{k\sigma} + \hat{a}^{+}_{k\sigma}\, e^{i\omega_k t - ikr}\, \hat{e}^{*}_{k\sigma} \right) \times$$

$$\times \left(i\, \hat{a}_{k'\sigma'}\, e^{-i\omega'_k t + ik'r}\, \hat{e}_{k'\sigma'} - i\, \hat{a}^{+}_{k'\sigma'}\, e^{i\omega'_k t - ik'r}\, \hat{e}^{*}_{k'\sigma'} \right)$$

Terms involving $\int d^3\vec{r}\, e^{i\vec{r}(\vec{k}+\vec{k}')} \left(\sqrt{\frac{\omega_k}{\omega_{k'}}}\, i\hat{a}_{k\sigma}\, \hat{a}_{k'\sigma'} - \sqrt{\frac{\omega_{k'}}{\omega_k}}\, \hat{a}_{k\sigma}\, i\hat{a}_{k'\sigma'} \right)$

just cancel out $= 0$.

Whereas those giving $\int d^3\vec{r}\, e^{\pm i\vec{r}(\vec{k}-\vec{k}')} = V\delta_{\vec{k},\vec{k}'}$ add up.

$$\vec{J}_{spin} = \sum_{k\sigma\sigma'} \frac{\hbar}{4V} \cdot 2V \left(i\, \hat{a}_{k\sigma}\, \hat{a}^{+}_{k\sigma'}\, \hat{e}_{k\sigma} \times \hat{e}^{*}_{k\sigma'} - i\, \hat{a}^{+}_{k\sigma}\, \hat{a}_{k\sigma'}\, \hat{e}^{*}_{k\sigma} \times \hat{e}_{k\sigma'} \right)$$

$$\hat{a}_{k\sigma}\, \hat{a}^{+}_{k\sigma'} = \delta_{\sigma\sigma'} + \hat{a}^{+}_{k\sigma'}\, \hat{a}_{k\sigma}$$

$$\vec{J}_{spin} = \frac{i\hbar}{2} \sum_{k\sigma\sigma'} \left\{ \hat{a}^{+}_{k\sigma'}\, \hat{a}_{k\sigma}\, \hat{e}_{k\sigma} \times \hat{e}^{*}_{k\sigma'} + \hat{a}^{+}_{k\sigma}\, \hat{a}_{k\sigma'} \cdot \hat{e}_{k\sigma'} \times \hat{e}^{*}_{k\sigma} + \delta_{\sigma\sigma'}\, \hat{e}_{k\sigma} \times \hat{e}^{*}_{k\sigma'} \right\}$$

(can switch $\sigma \Leftrightarrow \sigma'$ here)

$$= i\hbar \sum_{k\sigma\sigma'} \left(\hat{a}^{+}_{k\sigma'}\, \hat{a}_{k\sigma} + \frac{1}{2}\delta_{\sigma\sigma'} \right) \hat{e}_{k\sigma} \times \hat{e}^{*}_{k\sigma'}$$

1) Linear polarization: say, $\hat{e}_{k1} = \hat{x}$, $\hat{e}_{k2} = \hat{y}$

$$\Rightarrow \hat{e}_{k1} \times \hat{e}^*_{k2} = \hat{z} = \frac{\vec{k}}{k} = \hat{k} = \hat{n}$$

for $\sigma \neq \sigma'$

direction of wave's propagation

$$\vec{J}_{spin} = i\hbar \sum_{\vec{k}} \hat{a}^+_{k1} \hat{a}_{k2} \underbrace{\hat{e}_{k1} \times \hat{e}^*_{k2}}_{\hat{k}} + \hat{a}^+_{k2} \hat{a}_{k1} \underbrace{\hat{e}_{k2} \times \hat{e}^*_{k1}}_{-\hat{k}}$$

$$= i\hbar \sum_{\vec{k}} \left(\hat{a}^+_{k1} \hat{a}_{k2} - \hat{a}^+_{k2} \hat{a}_{k1} \right) \hat{k}.$$ (off-diagonal operator)

2) Circular polarization: here $\hat{x} \times \hat{y} = \hat{z} = \hat{k}$ unit vector along wave $\vec{k}$.

$$\hat{e}_{k1} = \frac{1}{\sqrt{2}} (\hat{x} + i\hat{y}), \quad \hat{e}_{k2} = \frac{1}{\sqrt{2}} (\hat{x} - i\hat{y})$$

then $\hat{e}_{k1} \times \hat{e}^*_{k2} = \frac{1}{2} (\hat{x} + i\hat{y}) \times (\hat{x} - i\hat{y})^* =$

$$= \frac{1}{2} (\hat{x} + i\hat{y}) \times (\hat{x} + i\hat{y}) = 0$$

but $\hat{e}_{k1} \times \hat{e}^*_{k1} = \frac{1}{2} (\hat{x} + i\hat{y}) \times (\hat{x} - i\hat{y}) = -i\hat{z}$

$$\hat{e}_{k2} \times \hat{e}^*_{k2} = i\hat{z}$$

$$\vec{J}_{spin} = i\hbar \sum_{k} \left(\hat{a}^+_{k1} \hat{a}_{k1} + \frac{1}{2} \right) (-i\hat{k}) + \left(\hat{a}^+_{k2} \hat{a}_{k2} + \frac{1}{2} \right) (i\hat{k})$$

$$= \sum_{k} \hbar \hat{k} \left(\hat{a}^+_{k1} \hat{a}_{k1} - \hat{a}^+_{k2} \hat{a}_{k2} \right)$$

$\Rightarrow$ photons with $\sigma = 1$ carry spin $(+\hbar)$ along $\hat{k}$ — circular right-

$\sigma = 2$ $(-\hbar)$ along $\hat{k}$ — left-polarized